

Agent-based modeling of energy efficiency policies: Enhancing sustainable development and economic growth

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ABSTRACT

This research examines how energy efficiency measures impact macroeconomic results and solves an important research deficiency. An analysis of how energy efficiency policies impact macroeconomic dynamics occurs through agent-based modeling (ABM). This research evaluates two approaches to sustainable economic development by studying tax incentives as indirect measures and public research laboratory R&D investments as direct measures. The implemented research and development-focused energy efficiency policies create superior environmental benefits while preserving economic stability, according to research results. The study demonstrates why carefully planned government policies are essential for sustainable ecological development and economic stability. The research data provides crucial knowledge for government officials who need practical guidance for merging energy efficiency strategies into economic planning systems. This research expands knowledge about energy efficiency relationships between the environment and the economy while offering directions for additional study on the multiple connections between energy use and economic stability and environmental objectives.

1. Introduction

Environmental sustainability, together with economic and social sustainability, needs immediate action due to growing recognition in recent times. Geopolitical tensions, together with COVID-19 supply chain disruptions, have increased energy prices, which strongly affected the purchasing power of importing nations. Low-income families, together with small and medium-sized businesses, face disproportionate impacts, which leads to increased social inequality [1]. Multiple climate change indicators have enhanced public awareness both inside diplomatic communities and beyond their boundaries while receiving global recognition at Conference of the Parties (COP) meetings. These challenges demand immediate and fundamental solutions because of the current urgent situation.

The reduction of energy intensity serves as a fundamental strategic solution for both emission reduction and ongoing energy crisis management. The energy intensity ratio measuring GDP against energy consumption decreased because of modern technological advancements during the past few decades. The current global improvement level falls short of the yearly targets described in Sustainable Development Goal

7.3, according to Dwivedi et al. [2] and Jun et al. [3]. The International Energy Agency (IEA) reports that several significant world economies show no signs of improving their energy efficiency policies, thus causing progress towards better efficiency to slow down.

Global nations endorse energy intensity reductions because they simultaneously help fight climate change while safeguarding against future energy emergencies [4]. The main drivers of reduction are technological advancements inside industries (technological effect), and industry shifts toward energy-saving sectors (structural effect). The main driver of energy intensity reductions has been technological progress, which exceeds structural transformations in numerous countries, according to Aristizabal-H et al. [4] and IEA (2016). The measurement of energy efficiency becomes challenging when energy savings through the rebound effect result in increased consumption because the indirect nature of this phenomenon makes it difficult to track [5].

Analysis of energy efficiency policy interventions and their effects on energy consumption remains essential to handling energy security challenges and climate change matters. The research utilizes macroeconomic agent-based modeling (ABM) to perform a complete assessment of various energy efficiency policies. Governments use direct

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technology interventions alongside indirect financial tools, including tax subsidies and incentives, to fix market failures. The research analyzes mission-oriented policies which solve major societal problems by using systemic public initiatives [6]. These policies work to advance four specific energy-efficient breakthrough technologies, which include heat pumps alongside homogenization valves, mechanical vapour recompression, and anaerobic wastewater treatment.

The study points out that an entrepreneurial state needs to support dangerous R&D projects, which private companies tend to bypass because they lead to breakthrough innovations. The mission-oriented approach surpasses standard agent-based models because these models mainly study household adoption of energy technologies (Bhavnagar et al., 2021). The research investigates mission-oriented policies which impact energy efficiency through an analysis that addresses essential knowledge gaps about these elements.

This research contributes an important advancement since it helps explain the relationship between energy efficiency policies and macroeconomic systems. The new ABM system outperforms traditional macroeconomic models by reaching its intended goals. The model uses diverse agents which mimic realistic interactions between production investment and energy consumption. Multiple policy simulation tests conducted within this research help discover the best strategies to reach environmental and economic targets. The analysis demonstrates how different economic sectors and social groups react to policy changes, which leads to understanding fair policy results. Research laboratory policies demonstrate the necessity of ongoing policy backing to realize energy efficiency and economic sustainability objectives. The research delivers fundamental direction to policymakers who need to create enduring sustainable plans for forthcoming periods.

(Fig. 1) shows that energy intensity decreased dramatically between the 1990s and the 2020s when units fell from 130 to 75. The 1990s represented the maximum energy intensity point at 130 units. During the 2000s, energy intensity levels decreased from 130 units to 120 units, and they reached 105 units during the 2010s. Energy systems, along with policy implementation, resulted in a major reduction of energy intensity, which decreased from 2010 to the 2020s down to 75 units. Economic policies, together with technological progress and global decarbonization programs, have led to a decrease in these trends. The data points allow Chinese policymakers to develop better clean energy plans while supporting economic development through evaluations of

national green energy policies and clean technology deployments. Each decade in the figure is displayed using different colour codes: blue represents the 1990s while orange stands for the 2000s, and red represents the 2010s, followed by pink for the 2015s and purple for the 2020s.

The manuscript is organized as follows:

- Section 2 provides an overview of the literature review, identifying research gaps in energy efficiency, renewable energy integration, and the relationship between policy and macroeconomic outcomes.
- Section 3 presents the methodology, including the agent-based modeling (ABM) approach and the simulation of various energy policies.
- Section 4 discusses the results, detailing the impact of energy efficiency policies on energy consumption, carbon emissions, and economic growth across different policy scenarios.
- Section 5 concludes with a discussion on the implications of the findings, recommendations for policymakers, and directions for future research.

2. Literature review

Energy efficiency policies represent a core element of worldwide sustainability and economic development discourse, particularly during the current global challenges involving energy price increases, supply disruptions, and post-COVID economic transformations [7]. The countries that depend on imported energy resources experience heightened vulnerability because low-income populations and small to medium-sized businesses sustain most of these impacts. This situation exacerbates socioeconomic inequalities and calls for more effective energy efficiency strategies [8]. The analysis investigates the mutual relationship between macroeconomic systems and energy efficiency policies regarding their effects on environmental sustainability and social equity. The research review exposes essential knowledge gaps, so scientists must develop policy frameworks that meet sustainability targets without creating economic challenges in the present moment.

2.1. Understanding energy intensity and policy trends

Historical research shows that energy intensity provides crucial information about economic energy efficiency through its calculation of

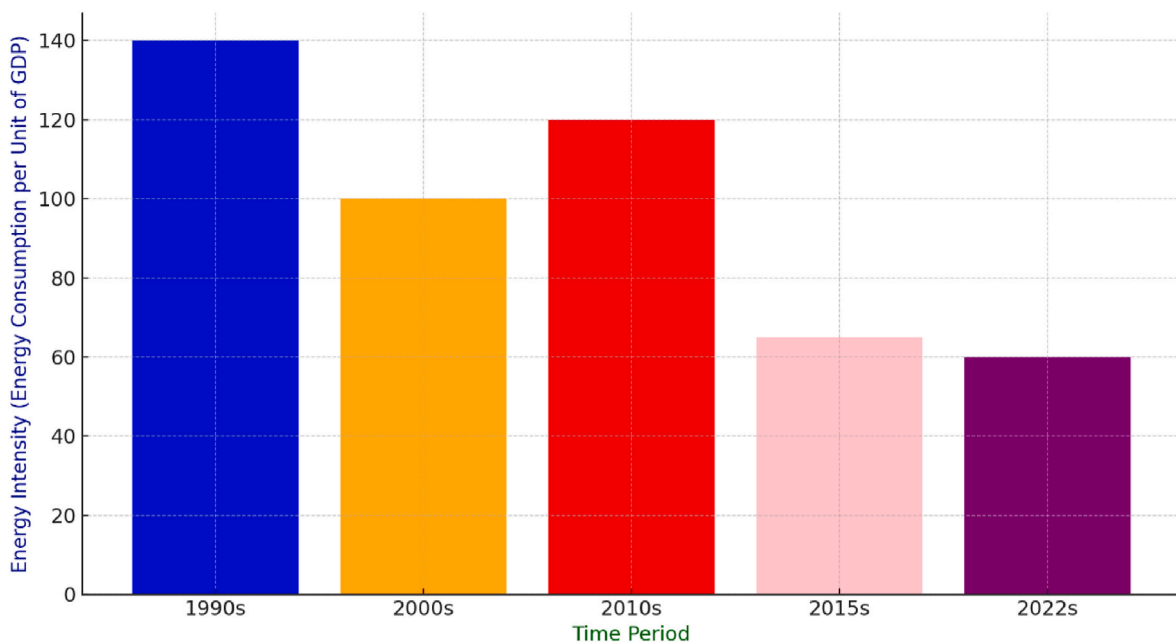


Fig. 1. Trends in energy intensity (energy consumption per unit of GDP) over the decades.

GDP divided by energy use [9]. The progress toward reduced energy intensity has slowed down, although technological advancements have produced these positive outcomes. The International Energy Agency (IEA) reports that major economies experienced a slowdown in energy efficiency because they lack sufficient energy efficiency policies, which prevents them from achieving Sustainable Development Goal (SDG) 7.3 for improving energy efficiency [10]. The pathway toward energy efficiency development from 1970 until 2020 is depicted in (Fig. 2), which illustrates how effective policies delivered these efficiency milestones (see Fig. 3).

Research has shown that two main factors contribute to energy intensity reduction:

1. The advancement of technologies leads to better industrial energy efficiency [11].
2. Service-oriented economic transitions serve to decrease total energy use throughout the economy. Technology advances function as the main driver, yet the rebound effect poses difficulties because energy savings trigger higher consumption rates [12]. The achievement of sustainable energy reductions depends on policies which unite technological progress with regulatory mechanisms [13].

2.2. Policy interventions for energy efficiency

The energy efficiency gap receives attention through the implementation of direct and indirect policy measures.

Direct Technological Interventions: Public investment in R&D for energy-efficient technologies.

The reduction of energy consumption through economic policy instruments includes taxes together with subsidies and incentives.

The growing popularity of mission-oriented policies has emerged because these policies tackle major social challenges by implementing systemic public intervention programs. The policies work toward developing heat pumps while targeting fundamental energy security concerns. Agent-based modeling (ABM) demonstrates value in evaluating policies through its capability to simulate how different sectors behave regarding energy consumption and investments.

2.3. The role of public research laboratories in energy efficiency

Public research laboratories now serve as primary institutions for advancing energy efficiency. Through their entrepreneurial nature, these institutions execute R&D projects with high risks, which private companies usually cannot afford because of financial limitations [14]. The combination of government-funded R&D programs under mission-oriented innovation policies proves to be the most effective approach for advancing energy efficiency targets. According to Wu et al.

[15], sustainable energy outcomes require both long-term investments in R&D and short-term measures such as subsidies and taxes.

2.4. Literature gaps and future research directions

Despite the extensive research on energy efficiency policies, several gaps remain:

Research about energy efficiency policies mainly concentrates on single nations and regions, thus creating a need for comparative studies across different countries. Future research needs to evaluate the operation of energy efficiency policies between different nations while considering their distinct environmental factors and socioeconomic conditions [16].

- **Longitudinal Policy Impact:** Many studies focus on the short-term outcomes of energy efficiency policies. Researchers need to investigate the extended-duration effects that these policies create on energy consumption and economic productivity [17].

Existing studies mention the rebound effect but lack standardized methods to measure it. The key issue of the rebound effect, along with its effects on energy consumption [17], demands additional research.

The Summary (Table 1) of Literature Contributions serves as a valuable resource for combining critical research findings from selected studies that relate directly to this present investigation. The following discussion will decompose this subject within your research framework:

This literature review underscores the critical role that energy efficiency policies play in achieving sustainability and economic stability. While technological advancements are key drivers, policy interventions remain essential in bridging the energy efficiency gap. Future research should focus on comparative studies, the long-term impact of policies, and the rebound effect to create a more comprehensive understanding of energy efficiency dynamics.

3. Data and method

The research uses an Agent-Based Model (ABM) to study how energy efficiency policies affect macroeconomic systems. The selection of ABM occurs because it effectively models heterogeneous agent behaviours as well as their system-wide interactions. Traditional modeling approaches fall short because Agent-Based Modeling effectively includes complex non-linear system dynamics and feedback processes that help achieve accurate simulations of energy policy impacts on energy efficiency alongside economic growth and social equity outcomes.

This (Table 2) includes the **abbreviation, variable name/description, measurement unit, and data source** to enhance clarity in the manuscript.

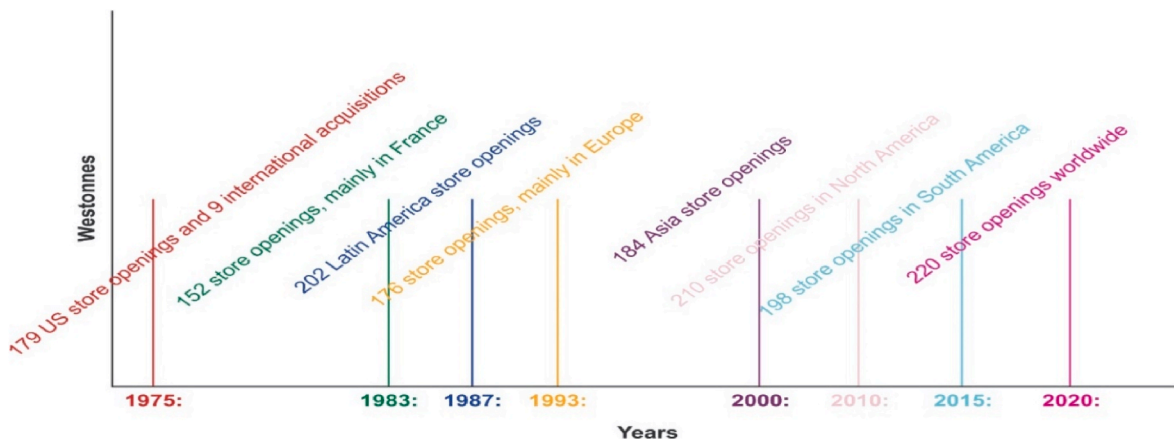


Fig. 2. Milestones in energy efficiency development over time (1970–2020).

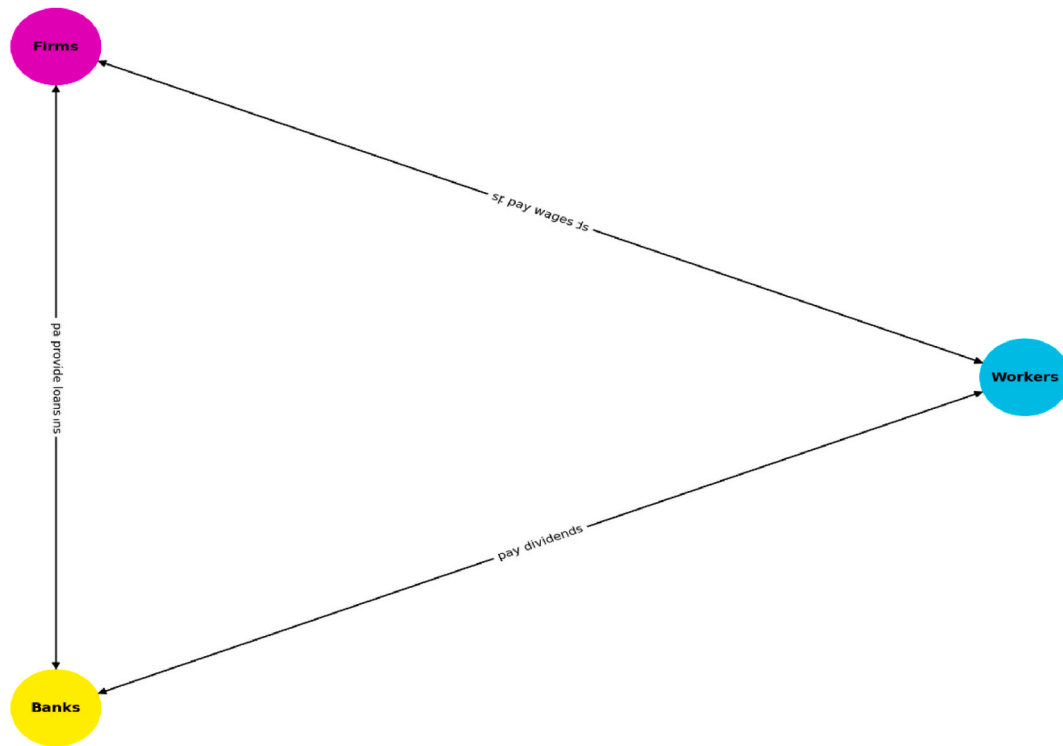


Fig. 3. Components of the model structure.

Table 1
Summary of literature contributions.

Study	Key Findings	Relevance to Current Research
[18]	Discussed the global challenges posed by rising energy costs and their impact on low-income communities.	Highlights the need for comprehensive energy efficiency strategies to address socioeconomic inequalities.
[19]	Identified technological advancements as the primary driver of energy intensity reduction.	Supports the argument that energy-efficient technologies are essential for achieving SDG 7.3.
[20]	Explored the role of mission-oriented policies in promoting energy efficiency.	Aligns with the current research focus on government intervention to support energy-efficient technologies like heat pumps.
[21]	Identified the need for longitudinal studies on the impact of energy efficiency policies.	Calls for further research into long-term policy impacts, aligning with the gaps identified in this study.

3.1. Theoretical framework

This study employs **agent-based modeling (ABM)** to analyze the impact of energy efficiency policies on macroeconomic dynamics. Traditional macroeconomic models often fail to capture the heterogeneity of individual agents (households, firms, policymakers) and their interactions in complex adaptive systems [17]. Unlike equilibrium-based models, **ABM provides a bottom-up approach**, allowing the simulation of diverse decision-making behaviors and policy responses.

We build upon the **energy efficiency-economic growth nexus**, where energy efficiency improvements impact GDP growth through direct (technology adoption) and indirect (policy-driven) channels. Our framework integrates **direct technological interventions** (e.g., R&D investments) and **indirect policy measures** (e.g., subsidies, carbon taxes) to evaluate their effectiveness in reducing energy intensity.

The model in question is a complex and comprehensive “disequilibrium,” stock-flow consistent, agent-based model with a wide range of

Table 2
Data description and sources.

Abbreviation	Variable Name/Description	Measurement Unit	Data Source
GDP	Gross Domestic Product	USD (constant prices)	World Bank, IMF
EE	Energy Efficiency	Energy use per GDP unit	IEA, National Energy Reports
EI	Energy Intensity	MJ/USD (MegaJoules per USD)	IEA, UN Energy Statistics
CO2	Carbon Dioxide Emissions	Metric tons per capita	IPCC, World Bank
RE	Renewable Energy Share	% of total energy mix	IRENA, EIA
TAX	Energy Tax Rate	% of energy cost	National Tax Reports
SUB	Energy Subsidies	USD per unit energy	Government Reports, IEA
R&D	Research & Development Spending	% of GDP	OECD, National Research Institutes
LAB	Labor Productivity	Output per worker	National Labor Statistics, ILO
IND	Industrial Output	USD (constant prices)	World Bank, National Economic Surveys
POL	Policy Intervention Indicator	Dummy (0 = No, 1 = Yes)	Policy Frameworks, Government Reports
INF	Inflation Rate	% annual rate	World Bank, IMF, National Banks
UNEMP	Unemployment Rate	% of workforce	National Labor Statistics, ILO
CON	Energy Consumption	MWh (Megawatt hours)	National Energy Statistics, IEA

Note: The agent-based model simulations present variables with their abbreviations and descriptions, measurement units, and data sources. The analysis relies on data from internationally recognized organizations and national reports, which present reliable information for assessing energy efficiency policies alongside their macroeconomic effects.

participants, including workers, banks, and firms, all subject to heuristic rules that determine how they make decisions. Crucially, the labor-augmented version of the K + S model, as presented by Ref. [22,23], is built upon this approach. Notably, it broadens its purview by adding energy as a further input to production, recognizing the critical role that energy dynamics play in forming economic systems.

This model implements technological progress through the DSK framework, which Lambert and their associates have extensively researched across 2018, 2019, 2020, and subsequent years. The dynamic structure of this model shows how this transformative power changes labour efficiency levels while significantly impacting energy usage effectiveness. Multiple elements, including market requirements and price relations between labour and energy and technical possibilities, determine both the origin and development of technological advancements.

3.2. Empirical approach and model justification

The K + S framework serves as the base for this study because it combines energy efficiency with economic development. The model includes the effects of technology and organizational structure on decreasing energy intensity. The adoption of energy-efficient technologies like heat pumps, technological advancements, and sectoral shifts function as primary elements that drive energy efficiency. Through its simulation capabilities, the ABM framework enables researchers to test various energy efficiency policies that include direct technology changes and indirect strategies of taxation and financial support systems [16].

The research benefits from using ABM since this methodology enables the simulation of agent decisions that operate independently and dynamically. Agents, including households, firms, and governments, show responses to different policies that depend on their specific characteristics within the economic environment. The approach provides enhanced insights into how diverse policy measures affect energy efficiency adoption rates and their effects on economic development [14].

3.3. Model calibration and sensitivity analysis

Historical data about energy use, together with GDP growth and technological advancement processes, enable the ABM calibration. The model requires inputs that allow researchers to correct its representation of actual agent activities. The model undergoes sensitivity testing to validate its predictive accuracy across different assumptions, specifically about energy prices, policy measures, and technological developments. The analysis allows the model to test various policy scenarios against their effects on energy consumption while verifying economic performance predictions.

3.4. Policy simulation and scenario testing

The study applies multiple energy efficiency policy simulations to determine their expected results. The policy measures involve direct technological investments in energy-efficient technology research and development as well as indirect economic instruments, such as incentives, carbon taxes, and subsidies for clean energy adoption. The model executes simulations to determine how various policy measures would accomplish reduced energy intensity and sustainable economic expansion. The simulation results show which policy strategies work best to enhance energy efficiency and reduce economic barriers during a low-carbon economy transition.

3.5. Optimizing production line efficiency

Manufacturing capital goods requires machines that depend on both labor and energy, similar to the production facilities for consumption goods. The technological capabilities of capital goods firms are determined by four key factors: machine labor productivity, manufacturing

process efficiency, and the energy utilization efficiency of the produced machines. These factors reflect both the operational efficiency of the machines and the inherent energy and labor efficiency of the firm's manufacturing processes.

The firm's unit production cost is defined by the average monetary salary it pays and the price of energy.

$$c_{i,t} = \frac{w_{i,t}}{B_{i,t}^L} + \frac{p_t^{en}}{B_{i,t}^{EF}} \quad (1)$$

The two terms are the firm's unit labor and energy expenses when creating the vintage. The price of the machine is then calculated by applying a predetermined mark-up pricing rule to the unit cost of production.

$$p_{i,t} = (1 + \mu_1)c_{i,t}, \mu_1 > 0 \quad (2)$$

Consumer product manufacturing companies intentionally use labor, energy, and machines to create standardized items with consistent returns to scale. The organization prepares production plans by predicting market needs and establishing extra capacity for unexpected customer demand increases. These organizations use visionary methods to link their production schedules with projected market requirements, thus creating a flexible structure that adapts to changes in consumer demand. Production unit costs are influenced by machine quality, staff skill, average wage rates, and energy expenses.

$$c_{j,t} = \frac{w_{j,t}}{A_{j,t}^L} + \frac{p_t^{en}}{A_{j,t}^{EF}} \quad (3)$$

Where $\bar{A}_{j,t}^L$ and $\bar{A}_{j,t}^{EF}$ each company's normal labor efficiency and energy efficacy within their production processes are expressed. The analysis produces metrics by evaluating machinery capital stock for labour efficiency, energy efficacy, and employee skill levels.

These businesses use a strategic pricing methodology to establish the price of their consumer goods. This is figuring out how much production costs each unit and then tack on a variable markup. Because the markup is a variable, the company can adjust and react to factors like market dynamics, operating expenses, and strategic positioning. The dynamic approach in this pricing methodology helps consumption-good companies negotiate the market's intricacies and balance cost considerations and competitive price tactics.

$$p_{j,t} = (1 + \mu_{j,t})c_{j,t} \quad (4)$$

Companies evaluate average labor productivity and energy efficiency in manufacturing by examining machinery and personnel skill levels. They use a dynamic pricing system to determine manufacturing costs per unit and add a variable markup, allowing for adjustments to market dynamics, operational costs, and strategic positioning. This method enables consumer goods companies to manage market complexities while balancing cost considerations and competitive pricing strategies.

$$\mu_{j,t} = \mu_{j,t-1} \left(1 + v \frac{f_{j,t-1} - f_{j,t-2}}{f_{j,t-2}} \right), v > 0 \quad (5)$$

One characteristic of the consumption-good market that sets it apart is incomplete information, which indicates that customers wait to immediately switch to the producer that offers the best terms in the market dynamics (Rotenberg, 2008). It is found that market shares evolve according to a (quasi) replicator dynamics framework, in which a constant flux characterizes market behavior. In this dynamic environment, firms with higher levels of competition expand and take up a more significant market share, whereas firms with lower levels of competition contract and lose market share. This complex interplay of (quasi) replicator dynamics provides insights into the subtle changes in market shares over time by reflecting the continuing competition and strategic

moves within the market.

$$f_{j,t} = f_{j,t-1} \left(1 + \chi \frac{E_{j,t} - \bar{E}_t}{\bar{E}_t} \right) \chi > 0 \quad (6)$$

where $E_{j,t}$ is the attractiveness of firm j in dated t and $\bar{E}_t$ is the regular attractiveness in consumption-good segment in the similar dated. Attractiveness of practices, in try, be contingent on the value they charge and, on their skill, to evade circumstances of untaken request:

$$E_{j,t} = -\omega_1 p'_{j,t-1} - \omega_2 l'_{j,t-1} \omega_1, \omega_2 > 0 \quad (7)$$

where $p'_{j,t-1}$ and $l'_{j,t-1}$ are the specific regularized price and empty request stages.

3.6. The dynamics of technical change

The principal goods industry is the economy's starting point for applying technological progress. Businesses in this industry are motivated by the unwavering quest for profit maximization and market expansion, which pushes them to aggressively seek technological advancements through creative problem-solving techniques and imitating profitable business models. These companies allocate a specific portion () of their previous sales () to research and development (R&D) projects to accomplish this goal. This calculated R&D investment is vital for developing innovations that boost the industry's overall growth and competitiveness.

$$RD_{i,t} = vS_{i,t-1}, s > 0 \quad (8)$$

The essential research and development programs are led by professionals who are solely focused on this vital work. Strategic decisions are made based on a well-calculated average salary within the context of the business's activities. In addition to demonstrating the company's dedication to innovation, this financial commitment to R&D makes hiring a certain number of researchers possible. This customized hiring strategy guarantees that the labor force involved in these projects matches the allotted investment, enabling the best potential synergy between monetary resources and the human capital required to push technological boundaries.

$$L_{i,t}^{RD} = \frac{RD_{i,t}}{w_{i,t}} \quad (9)$$

Researchers are divided into groups based on the following parameter $\xi \in [0, 1]$: invention (IN) and imitation (IM).

$$IN_{i,t} = \xi L_{i,t}^{RD} \quad (10)$$

$$IM_{i,t} = (1 - \xi) L_{i,t}^{RD} \quad (11)$$

Drawing on the model offered by Ref. [24], the dynamic processes of innovation and imitation integrate intrinsic uncertainty, represented as a two-step stochastic system. The first crucial step is determining a firm's propensity for revolution or simulated using two distinct pulls from Bernoulli deliveries. The characteristics influencing these distributions are inextricably linked to organizations' search capacities (), resulting in a complicated interplay between exploration ability and penchant for originality or copying. Furthermore, the many researchers involved in these activities complicate the decision-making process.

$$\theta_{i,t}^{in} = 1 - e^{-\zeta_1 IN_{i,t}}, \zeta_1 > 0 \quad (12)$$

$$\theta_{i,t}^{im} = 1 - e^{-\zeta_2 IM_{i,t}}, \zeta_2 > 0 \quad (13)$$

Based on the DSK model (Z. [25]), businesses that proceed to the another stage of the innovation process acquire four new technical constants by pursuing this particular approach:

$$A_{i,tn}^L = A_{i,\tau}^L (1 + x_{A,i}^L) \quad (14)$$

$$B_{i,tn}^L = B_{i,\tau}^L (1 + x_{B,i}^L) \quad (15)$$

$$A_{i,tn}^{EF} = A_{i,\tau}^{EF} (1 + x_{A,i}^{EF}) \quad (16)$$

$$B_{i,tn}^{EF} = B_{i,\tau}^{EF} (1 + x_{B,i}^{EF}) \quad (17)$$

Firms draw from two independent distributions while exploring new technologies: and, where x_A^L and x_B^L are random draws from distributions defined over the support. In the same way, they are separate draws from support-filled distributions. This dynamic suggests businesses are negotiating a four-dimensional technological area with their current technology at its core. According to Dose (1988), the range of each distribution $[\xi_1^{EF}, \xi_2^{EF}]$ dictates the variety of technical opportunities along the corresponding dimension. These distributions' bottom limit is negative, replicating the natural trial-and-error process of technological research and permitting the investigation of less-than-ideal technologies.

Businesses can imitate a competitor's technology when they successfully advance to the second stage of the imitative process. Notably, if competitors are technologically close to you, there is a greater chance of imitation. The following is the definition of the exact technological distance ($Beta(\alpha^{EF}, \beta^{EF})$) that separates two firms (and):

To elaborate on the idea, this implies that companies traverse a complex terrain, investigating technological avenues beyond their present technical condition. Random draws from distributions define the possible opportunities that impact each dimension. This acknowledges the possibility of discovering inferior and superior technologies, reflecting technological discovery's dynamic and uncertain character.

When it comes to imitation, closeness to technology affects the target selection. Businesses tend to copy the technology of technologically closer competitors, which indicates a strategic preference for accessibility and familiarity during the imitation process.

$$TD_{i,g} = \sqrt{\left(\frac{A_{g,\tau}^L - A_{i,\tau}^L}{A_{i,\tau}^L} \right)^2 + \left(\frac{B_{g,\tau}^L - B_{i,\tau}^L}{B_{i,\tau}^L} \right)^2 + \left(\frac{A_{g,\tau}^{EF} - A_{i,\tau}^{EF}}{A_{i,\tau}^{EF}} \right)^2 + \left(\frac{B_{g,\tau}^{EF} - B_{i,\tau}^{EF}}{B_{i,\tau}^{EF}} \right)^2} \quad (18)$$

Adopting a new technology is simple when it outperforms the existing one in all areas. However, given trade-offs, judgments are made based on a thorough examination of the projected success of devices using various technologies. This evaluation follows principles similar to those consumer goods businesses use to pick machine providers. Aside from technological benefits, the decision-making process includes a practical examination of market factors, emphasizing the importance of predicted success and efficiency in creating the technology adoption landscape.

$$\min_{i \in I_{j,t} \subseteq F_t^1} \{p_{i,t} + bc_{i,t}\} \quad (19)$$

Where parameter denotes the payback period b an essential factor in consumption-good companies' investment plans. A further explanation of this parameter is given later in the text. The optimization method is limited to a particular subset of capital-good enterprises due to the imperfect information in the market for capital goods. This subset is distinguished by each consumer-good firm's awareness of ($I_{j,t}$).

The index ji^* denotes the process-selected capital firm. As such, the model includes localized interactions between suppliers and customers that fall under this subgroup.

Capital firms use a set of rules when deciding what kind of machine to create. These rules are as follows:

$$\min_h \{p_{i,t}^h + b\bar{c}_{i,t}^h\}, h = in, im, \tau \quad (20)$$

Where $p_{i,t}^h$ and $\bar{c}_{i,t}^h$ are respectfully When making decisions, consumption-good companies consider average unit production costs and pricing, especially regarding machines linked to different technologies.

This decision rule adds to the model a dynamic that Hicks first proposed in 1932 and calls for directed technological change. According to this paradigm, the relative costs of labor and energy play a significant role in determining which machine to build, in addition to technical constraints. More specifically, the comparative pricing dynamics of labor and energy determine how much weight each technological parameter gives. For example, technological characteristics related to energy efficiency become more important if there is an increase in the relative cost of energy. Businesses will probably be encouraged to adopt technologies with higher levels of energy efficiency as a result of this change in relative priority. This decision-making paradigm captures the complex interaction between economic and technological elements, adjusting to the dynamic changes in market input prices.

3.7. Dissemination of technological innovations

Firm-level investments in the consumer goods sector are critical to generating economic technology breakthroughs. Consumer goods firms invest strategically in new machinery and influence the adoption of novel manufacturing practices in the capital goods industry. This bidirectional link emphasizes the critical role of consumption-goods businesses in fueling and igniting broader economic trends.

Businesses engage in investment activities with two primary goals: (i) growing their capital stock and (ii) replacing antiquated equipment with newer models. Keeping an eye on the first goal, consumption-good companies use a linear function connected to their production targets to determine their ideal capital stock ($K_{j,t}^d$). Businesses make strategic investments to enable the augmentation of capital stock ($El_{j,t}$) when it is below the targeted level. The following factors precisely define the size of these intended expansionary investments ($El_{j,t}$):

$$El_{j,t} = K_{j,t}^d - K_{j,t} \quad (21)$$

A payback schedule, which involves companies methodically balancing the advantages of technological $Sl_{j,t}$ expenditures against the expenses of purchasing new equipment, serves as guidance. This assessment is carried out with the particular period in mind that they hope to recover their investment (b). To elaborate, the choice to replace the vintage τ machine with a new one is made based on a variety of factors within the range of machine vintages that the company has at any one moment, including:

$$\frac{p_t^{ii*}}{c_{j,t}^{\tau} - c_{j,t}^{ii*}} \leq b \quad (22)$$

where p_t^{ii*} is the cost of the new machine and $c_{j,t}^{ii*}$ and $c_{j,t}^{\tau}$ respectively, the unit costs of production for the old and new machines. The sum of the replacement and expansionary investments makes up the overall investment.

3.8. The role of government in society

The government collects taxes on bank and company earnings and profits, allots funds for unemployment insurance, purchases consumer products for the general public, gets involved in bank bailouts, and sets minimum wage requirements. The government actively implements energy efficiency regulations in the following sections of the paper, especially in the policy experiments that are described. Considering these various budgetary endeavors, the public deficit is defined as:

$$Def_t = (w_t^u U_t + G_t^c - tr^{\pi} \Pi_t - tr^{In} In_t) + r^{deb} Deb_{t-1} + Bail_t - ET_t + G_t^{ef} \quad (23)$$

where U_t and, respectively G_t^c , indicate the in the baseline scenario, both energy tax revenues and public spending on energy efficacy programs are zero. where is the total number of unemployed people, the expenditures on public consumption tr^{π} and tr^{In} the tax rate on incomes and pay, and the total profits and profits, the interest paid on the debt, and the spending for bank bailouts.

3.9. The economics of energy costs

International market dynamics determine a predetermined price that influences energy purchases from worldwide sources. In line with other predictions for the future, as seen in several estimates before the current energy crisis [26], we assume that the price of energy will grow at a moderately positive pace over time. To express this assumption, we propose a constant growth rate that is the same throughout time:

To elaborate, the premise of a constant growth rate suggests that energy prices will gradually rise, a trend commonly recognized in forecasting models and studies of the energy industry. It is necessary to start with this assumption to comprehend and project the dynamics of energy prices within the modeled economic framework.

$$p_t^{en} = p_{t-1}^{en} (1 + \xi^{en}) \quad (24)$$

Where $\xi^{en} > 0$

Energy prices are rising, and this trend is based on several essential presumptions. First and foremost, a crucial element is the predominance of fossil fuels in energy production. According to projections from the International Energy Agency's World Energy Outlook 2022 [27], natural gas creation is probable to top in the upcoming ten years, while oil production is predicted to peak in 2035 [28]. Despite these predictions about depleting fossil fuel stocks, the IEA's Stated Policies Scenario predicts that by 2050, the world's share of energy derived from fossil fuels will still surpass 60 %.

Second, the examined economic model shows a smaller economy primarily dependent on energy imports from other countries. As a result, determining the price of energy on global markets is generally immune to variations in the demand for power within a country, highlighting the significant impact of external forces [29].

A crucial component is the complex relationships between energy costs, decarbonization programs, and energy efficiency. Our model adopts a unique approach, even though it acknowledges that activities like installing renewable energy infrastructure can lead to efficiency gains—as demonstrated by Ref. [30] analysis of energy savings associated with solar windows (2022). Energy efficiency innovation is handled independently of the carbon satisfied of the energy consumed by trade enterprises in our modeling framework. Essentially, this innovation neither promotes nor facilitates the broad use of low-carbon energy sources. This complex viewpoint acknowledges the possible effects of energy-related advances within the economic environment under discussion and is essential to our modeling methodology. The (Table 3) of Conventional Facts and Empirical Studies provides a necessary structure to understand the overall relationship between energy efficiency and technological innovation alongside macroeconomic activity. The combination of essential research results from numerous studies enables us to comprehend their impact on our current study better (see Table 4).

4. Empirical grounding: verifying simulated outcomes

Computer simulations become essential because the complex model lacks direct analytical solutions to study micro-macro coevolution processes. Our analysis depends on extensive Monte Carlo experiments that perform 100 simulation runs. The simulations run for 100 periods of initial warm-up followed by subsequent active simulation periods that

Table 3

Model precision: Replicating key stylized facts.

Conventional facts	Empirical studies (among others)
Energy cost share heterogeneity across firms	[31,32]
Macroeconomic stylized facts:	
Energy efficiency heterogeneity across firms	(J. [33,34])
Self-sufficient growth with persistent fluctuations	(W. [35]); (W. [36])
Fat-tailed GDP growth rate distribution	[37,38];
Cyclically-driven energy demand	[39]
GDP, feeding and venture instability	(Y. [40]);
Cross-correlation of macro variables	[41]; (Y. [42])
Recession duration exponentially distributed	[43];
Low energy/labor cost ratio	[44,45]
Cyclically-dependent R&D spending	[44]
Policy-driven technological innovation for energy efficiency	[46,47]

Table 4

Summary statistics covering 100 Monte Carlo (MC) runs are given for a subset of the variables. “Like.” indicates probability. A crisis is characterized by a one-time 3 % decline in GDP that does not fully recover over the following three periods. The percentage of businesses engaging in these activities is reflected in the likelihood of innovation and imitation.

Variable	MC Normal	MC St. Dev.
Likelihood of Imitation	0.151	0.019
Likelihood of Innovation	0.202	0.012
Likelihood of Crisis	0.102	0.065
GDP Volatility	0.063	0.031
GDP Growth	0.025	0.004
Energy/Labor Cost Ratio	0.124	0.057
Energy Intensity Growth	−0.009	0.009
Energy Efficiency Growth	0.009	0.01
Energy Demand Growth	0.018	0.012
Productivity Growth	0.022	0.002

extend for 400 periods. The model runs through 160 time periods to validate its performance through stylized fact comparison before executing 240 time periods of policy scenario simulation from 2020 to 2080. The analysis foundation stems from these simulations along with the indirect calibration method, which serves as the analytical foundation.

4.1. Unveiling the Model's real-world efficacy

The model successfully duplicates economic changes from short to long durations while maintaining the capability to reproduce stylized information through different variables. During the 100-run Monte Carlo simulation, the model produces endogenous growth along with business cycle fluctuations and periodic crises, as demonstrated in Table 2. The simulated results correspond to empirical research about business cycles that demonstrate both growth and opposite growth patterns.

Energy intensity decreases as GDP growth increases according to negative correlation patterns that match the observed empirical data.

The model incorporates behaviour patterns of innovation and imitation to drive internal energy efficiency and labour efficiency growth. The growth in energy efficiency produces declining energy intensity patterns that match observed real-world trends.

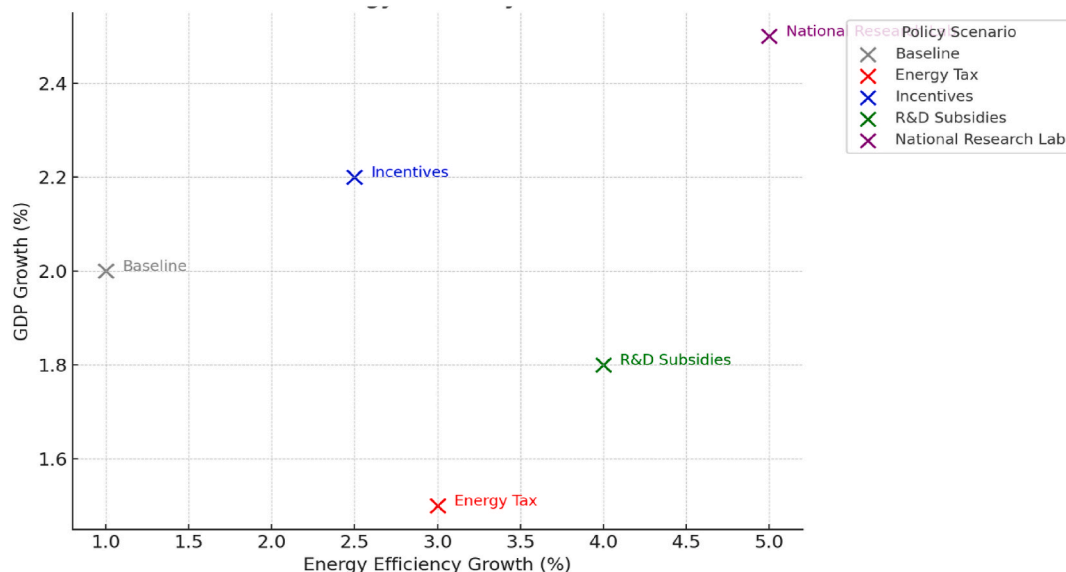
The Granger causality tests verify the existence of macroeconomic development rebound effects, thus demonstrating the model's ability to analyze dynamic feedback relationships between variables.

4.2. Descriptive statistics

This (Fig. 4) illustrates the correlation between **energy efficiency growth** and **GDP growth** across different time periods. The data show how improvements in **energy efficiency** coincide with periods of **economic growth**, with clear trends in energy conservation measures aligning with economic expansions. The figure helps to highlight how **policy interventions**, technological advancements, and energy efficiency strategies have influenced **economic growth** in each phase, especially in the context of sustainability and decarbonization policies.

Furthermore, the existence of “beta convergence” between Monte Carlo (MC) runs at the energy intensity level, which is an important finding. This effect implies that simulations that begin with greater energy intensity levels eventually see a more noticeable decrease in energy intensity (Fig. 4). This is consistent with empirical data showing a “beta convergence” pattern in energy intensity levels across nations [48]. This suggests there may be diminishing returns to reducing energy strength; it may be calmer to do so when energy strength is initially tall and more challenging as it approaches lower values.

Additionally, the model illustrates aspects of induced innovation related to energy efficiency, which align with results from other empirical investigations [49]. In particular, the model shows that the

**Fig. 4.** Relationship between energy efficiency growth and GDP growth over time.

economy's development and diffusion of extremely energy-efficient technologies are either strengthened or weakened by changes in the growth rate of energy prices. This relationship can be expressed as follows: for every 1 % increase in the growth rate of energy prices, there will be a corresponding 0.6 % increase in the growth of energy efficiency and a corresponding 0.65 % decrease in the growth rate of energy intensity.

Finally, at the micro level, the model exhibits notable variation in energy efficiency and the percentage of energy expenses across enterprises in both sectors, consistent with actual results [50] (see Fig. 5). This result emphasizes how the modeled economic framework realistically captures the heterogeneous landscape of energy-related attributes at the firm level.

4.3. Experimental approaches to policy analysis

Our research uses scenario analysis to carefully consider the effects of different policies intended to improve the system's energy efficiency. Section 4.1 explains these policies, and Section 4.2 provides a detailed overview of the associated simulation results. Section 4.3 presents findings from other experiments, extending our investigation and providing a comprehensive viewpoint outside of the main policy trials.

In our assessment, we rigorously test four different policies by contrasting the average results obtained from the baseline scenario (described in Section 3) with those obtained from the policy scenario regarding energy efficiency and macroeconomic dynamics. The idea of the State correcting market failings serves as the basis for three of these policies: the energy tax, incentives encouraging the purchase of extremely energy-efficient machinery, and research and development subsidies. Simultaneously, the National Research Lab, the fourth policy, strategically reflects the mission-driven policies typical of the business State in its crucial function of "creating and shaping markets."

Crucially, the settings of the policy situations are carefully designed to promote an environment favorable to an impartial and equitable comparison of the various policy instruments. This careful adjustment highlights the strength of our research, offering complex perspectives on

the efficacy and ramifications of every policy direction investigated, going beyond the scope of the main policy trials.

4.3.1. Deciphering policy frameworks

Policy I revolves around the deliberate implementation of an energy tax strategically designed to elevate the overall cost of energy. The government's rationale behind this fiscal maneuver is grounded in the expectation that such a policy will serve as a potent catalyst, compelling businesses to innovate and adopt more energy-efficient production techniques. Within the experimental framework of this policy, firms grapple with a discernible increase in the effective price of energy, denoted by a multiplier of $1 + \tau \cdot \text{en}$, where $\text{trend} \in (0,1)$ meticulously signifies the energy tax rate set by the management.

Policy II represents an organized initiative to boost energy efficiency through direct financial incentives for Consumer goods firms to obtain advanced energy-efficient equipment. The policy offers a refund that equals $q\%$ of purchase costs to firms who buy machines with the A (++++) certification. The A (++++) label demonstrates that the machine achieves higher efficiency standards than the ones that the government periodically updates. The "A (++++) label" receives multiple benefits that remain active for the duration of k periods.

The strategic incentives control important aspects of firm decision processes in both manufacturing and consumption industries. The proactive cost reduction of A^ (++++) machines within the consumption sector functions as an effective motivator to boost investments into advanced technological machines. The policy stands as a vital element that will accelerate the adoption of high-energy-efficient technologies throughout different production levels. The incentives create a transformative effect on the decision-making processes for machine suppliers and replacement investments while changing the strategic approaches of this policy implementation.

$$\min_{i \in I_t, \subseteq F_t^1} \{p_{i,t}(1 - q_{i,t}) + bc_{i,t}\} \quad (25)$$

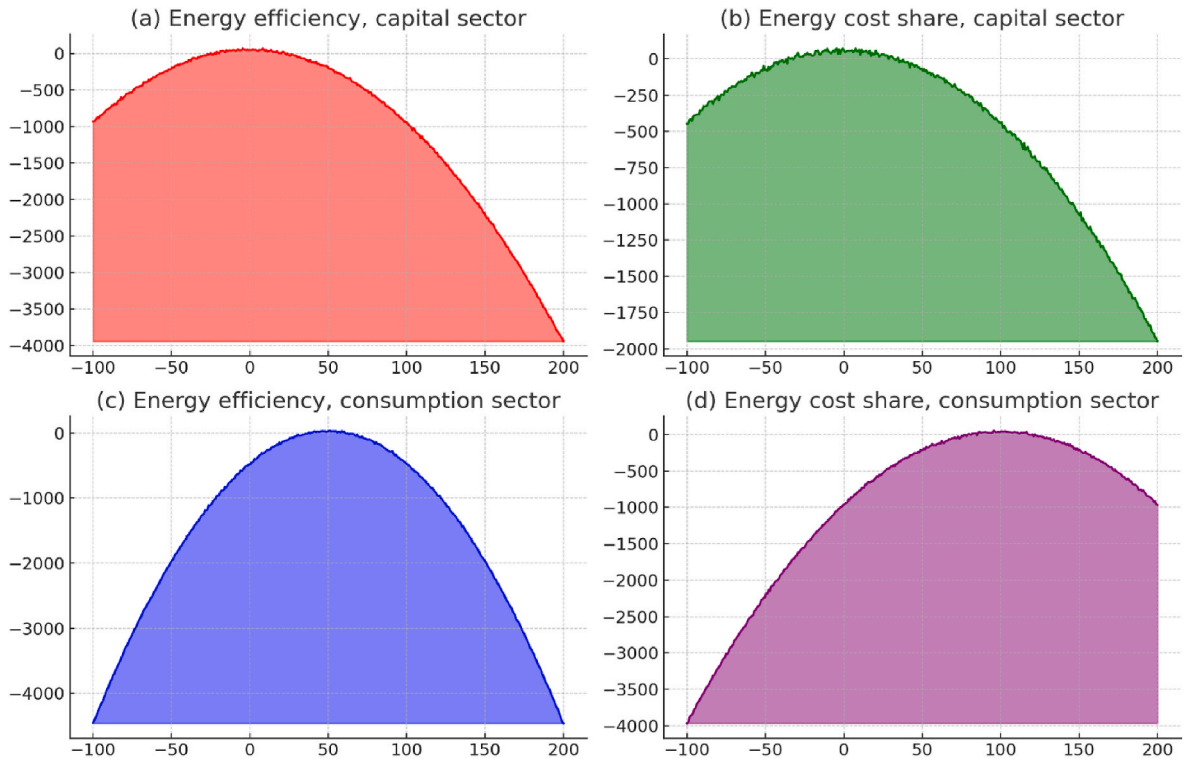


Fig. 5. "Variability in energy efficiency and costs: A sectoral analysis of capital and consumption firms".

$$\frac{p_t^*(1 - q_{i,t})}{(c_{j,t}^* - c_{j,t}^*)} \leq b \quad (26)$$

When capital-good firm i produces a machine with the classification, then the variable q obtains value 1; otherwise, it remains at 0. Within the capital sector, the existing incentives motivate firms to transform their technological methods while driving them to specialize in machine production. Capital-good firms transform their strategies because they see great potential in the rapidly growing market for advanced technologies. Firms in the machine sector use a weighted calculation method which combines discounted and undiscounted price values to determine their valuations.

The variable plays a critical role in determining capital-good company decisions during their decision-making processes. The A^+ (+++) designation earned by a machine leads to the addition of incentive value q in pricing decisions because it represents a commitment to high energy efficiency. When machines do not qualify for the category, there is no q included in the pricing mechanism, which demonstrates how incentives guide firm technology decisions.

Specialization decisions of firms in machine production become a central story element within the dynamic capital sector environment. This strategic transformation represents an active opportunity-seeking behaviour which results from the increasing market demand for advanced energy-efficient technologies. Businesses implement a weighted average method that combines discounted and undiscounted prices to help them move through market changes and achieve superior competitive positions. The dual consideration of pricing demonstrates sophisticated strategic elements present in capital industry decision-making when policy frameworks offer incentives.

$$\min_h \{ p_{i,t}^h (1 - \bar{\omega}) + p_{i,t}^h (1 - q_{i,t}) \bar{\omega} + b \bar{c}_{i,t}^h \}, h = in, im, \tau \quad (27)$$

Our model relies heavily on the $\bar{\omega}$ parameter because it defines how much weight reduced prices should carry. The extent of time firms predict they will benefit from incentives directly influences their willingness to modify technology plans. The average mortality rate in the capital sector shows a negative correlation pattern with the $\bar{\omega}$ value. A higher average mortality rate in the capital sector reduces the

importance of discounted prices during decision-making, thus demonstrating the complex relationship between economic elements and policy results.

Policy III, centered on Research and Development (R&D) subsidies, stands as a pivotal instrument in the government's strategy to invigorate technological progress. Through the allocation of financial resources to capital firms, the government aims to stimulate and amplify their research activities. This strategic infusion of funds is envisioned to act as a catalyst, fostering not only sector-specific advancements but also triggering a ripple effect that positively influences the broader landscape of technological innovation.

This (Fig. 6) shows the cumulative GDP growth from the 1990s to the 2020s, with key periods marked by economic expansion and periods of policy-driven energy efficiency improvements. The (Fig. 6) demonstrates how GDP growth has evolved alongside the implementation of energy efficiency policies and the adoption of sustainable technologies. It underscores the role of energy policies in sustaining economic growth while promoting energy conservation, with visible patterns of energy use optimization coinciding with increased economic activity (see Fig. 7).

The operationalization of this policy is methodical. In each retro, the management meticulously determines the entire economical for R&D supports RDS, expressing it as a fractional proportion of the nominal Gross Domestic Product (GDP). The planned approach leads to financial resource adaptations that maintain alignment with economic circumstances and the technological evolution of the present landscape.

$$RDS_t = \zeta GDP_{t-1}^{nom}, \zeta \in (0, 1) \quad (28)$$

This scenario is also repeated with the whole money distributed across all enterprises according to their past R&D spending.

$$RDS_{i,t} = RDS_t \frac{RD_{i,t-1}}{\sum_i RD_{i,t-1}} \quad (29)$$

In the III-bis version of the scenario, the government adopts a different funding strategy. Private companies obtain government funding through a specific Research and Development (R&D) project call, which provides resources to organizations demonstrating exceptional

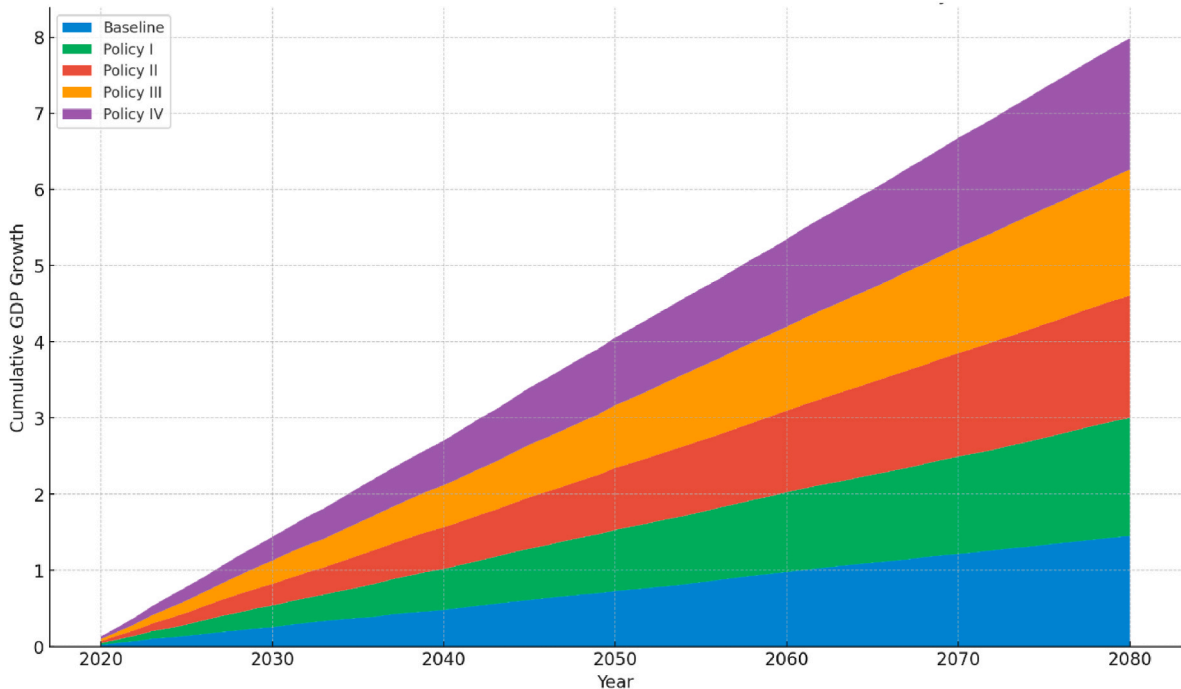


Fig. 6. Cumulative GDP growth over time and its relationship to energy efficiency policies.

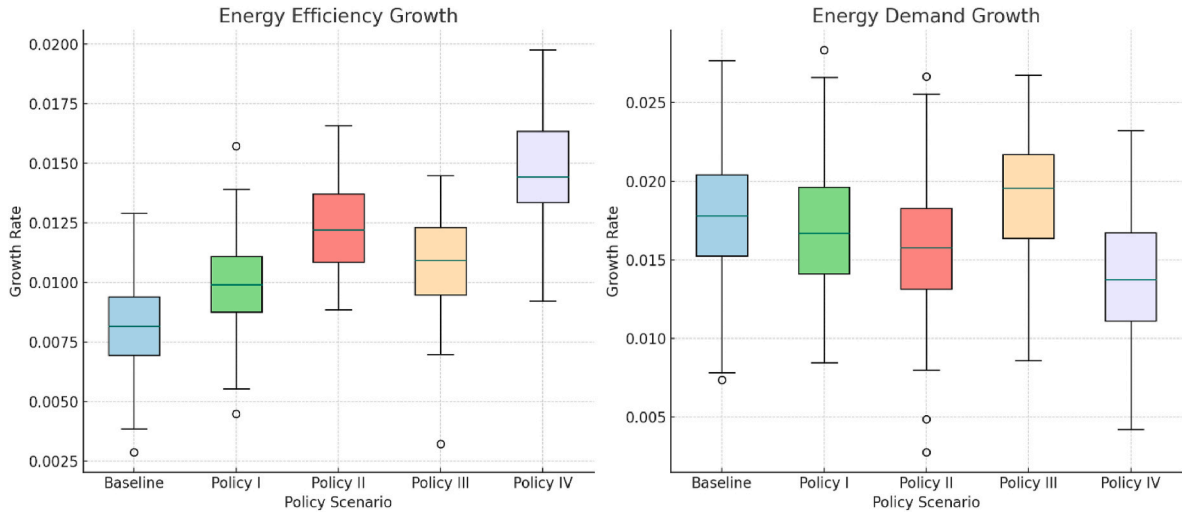


Fig. 7. Shows production and energy efficiency in the capital and consumption sectors in 2080. The data are expressed as means for no policy scenario. 100 MC cycles.

innovation or innovative projects. The strategic approach focuses money on businesses that demonstrate outstanding potential or have proven their technological dominance throughout history.

The framework enables the reproduction of exclusion through government R&D subsidy distribution, which targets superior technology companies, specifically in the area of energy efficiency. Subsidies allocation from the government depends on previous research and development spending by each company, linking past innovation investment to present funding opportunities. This targeted allocation method aims to reward and encourage companies that have continuously demonstrated a forward-thinking mindset and a dedication to pushing the boundaries of technology in the field of energy efficiency.

$$RDS_{i,t} = RDS_i \frac{I_{i,t-1} RD_{i,t-1}}{\sum_i I_{i,t-1} RD_{i,t-1}} \quad (30)$$

Within this context, we introduce the notation B^{EF} to signify the average energy efficiency characterizing the production process. Simultaneously, A^{EF} is employed to denote the average energy efficiency pertaining to machines crafted within the capital sector. The correlation between these variables is articulated as follows:

$$I_{i,t-1} = \begin{cases} 1 & \text{if } A_{i,t-1}^{EF} + B_{i,t-1}^{EF} > \overline{A_{i,t-1}^{EF}} + \overline{B_{i,t-1}^{EF}} \\ 0 & \text{otherwise} \end{cases} \quad (31)$$

We use lessons from several empirical investigations, as demonstrated by the work of [51], to conclude this topic. We include an extra parameter for the subsidy in our model, which is set to one. This represents a detailed analysis, implying that the overall Research and Development (R&D) cost for the ITH capital company is made up of the conventional components plus the additional impact of this subsidy factor. Through the incorporation of these nuances, our approach recognizes the complex nature of R&D investment and the influence of subsidies as a crucial part of capital businesses' overall expenditure equation.

$$RD_{i,t} = vS_{i,t-1} + RDS_{i,t} \quad (32)$$

Policy IV introduced a policy IV and introduced the creation of a National Research Lab on Energy Efficiency. Through this creative project, the government invests funds in a specialized research institution that aims to increase our knowledge of energy efficiency. The lab has been given a clear scientific mission: to lead innovative research projects that aim to not only create new technology paradigms but also make small but significant changes. There is no doubt that the main goal

is to drastically reduce the environmental impact of energy consumption by revolutionizing energy efficiency.

This policy scenario is based on [52] explanation of the notion of an entrepreneurial state. The crux is that the government is actively involved in mission-driven projects, which are similar to those supported by major organizations like the U.S. Department of Energy, ARPA-E, and those that are part of Germany's Energiewende framework. These programs highlight how aggressively the government is directing innovation toward environmentally friendly energy methods.

The government's investment in a National Research Lab on Energy Efficiency is an example of how seriously it takes the promotion of game-changing discoveries. This is more than just financial assistance; it is a calculated investment in innovative research with the potential to transform technology and usher in a new era of energy efficiency. This policy demonstrates a thoughtful attempt to match national goals with creative projects and a forward-thinking strategy for tackling environmental issues via ground-breaking R&D.

The operationalization of this policy unfolds in the following manner. At the onset of each historical, the management delineates the economical earmarked for the nationwide study lab. This budget allocation is determined as a fraction, denoted as η , of the minimal Gross Domestic Product (GDP):

$$RD_{NL,t} = \eta GDP_{t-1}^{nom} \eta \in (0, 1) \quad (33)$$

The national research lab hires a precise number of researchers, ascertained by using the budget that has been allotted:

$$L_{NL,t}^{RD} = \frac{RD_{NL,t}}{w_{NL,t}} \quad (34)$$

where $w_{NL,t}$ is their wage.

Our paradigm is grounded in the workforce dynamics of the national research lab, which is in line with the ideas clarified in the literature on the entrepreneurial state. This specialized crew works on energy efficiency projects, taking on tasks that have an extremely low chance of accomplishment. These projects present an inherent problem since they involve potentially innovative ideas with a very low level of technological readiness and a great deal of uncertainty surrounding their possible commercial uses. The public sector bears the brunt of this amount of complexity and uncertainty, which tends to discourage private-sector investment. The public sector is tasked with taking on high-risk projects in order to lead the establishment of a revolutionary technological paradigm because of its significant financial resources and strategic vision.

It's critical to emphasize that these extreme breakthroughs rarely work well inside a single study endeavor due to their nature. Rather, it usually entails a series of trial-and-error activities. Research efforts that are accumulated over time become essential for negotiating the complexities of innovation, as studies like those by Ref. [53,54] illustrate.

The model displays a structured format to detect these refined features. It uses Bernoulli random sampling of the distribution to represent the stochastic probability that the national lab will achieve an energy efficiency radical innovation (EERI). The methodology includes all uncertainties associated with finding breakthroughs in advanced energy efficiency research.

$$\theta_t^{EERI} = 1 - e^{-\zeta 1_{NL} \sum_{k=p}^{k=t} I_{NL,k}^{RD}}, \zeta 1_{NL} > 0 \quad (35)$$

The research at the national laboratory carries a high risk due to a $\zeta 1$ parameter value below one. Historical research, through cumulative summation, reveals that innovation develops through repeated cycles. Implementing Energy Efficiency Radical Innovations (EERIs) produces major technological alterations that open new business possibilities. These concept-based developments transform both operational areas and business paths of companies that target energy efficiency. This approach recognizes the dynamic interplay between innovation and technological opportunity.

$$\xi 1_{p=t}^{EF} = \xi 1^{EF} + \theta_{p=t}^{EF}; \xi 2_{p=t}^{EF} = \xi 2^{EF} + \theta_{p=t}^{EF} \quad (36)$$

The degree of the shift represented is contingent on both the specific technological paradigm and the time under consideration. This highlights the concept that the impact on energy efficiency technological opportunities varies across diverse technological paradigms and evolves. This assumption is grounded in the recognition that not all technological paradigms carry equal weight, and as time progresses, these paradigms undergo a gradual process of diminishing influence. With this in mind, the shift introduced by each paradigm is expressed through the following model:

$$\begin{cases} \theta_{p=t}^{EF} = \text{Unif} \sim [\theta_{min}^{EF}, \theta_{max}^{EF}]; \theta_{min}^{EF} > 0 \text{ if } I \text{ discovered in } t \\ \theta_{p=t}^{EF} = (1 - \varphi) \theta_{p=t-1}^{EF}; \varphi > 0 \text{ otherwise} \end{cases} \quad (37)$$

The length of time since a paradigm's implementation determines its level of exhaustion. Adopting a new technology paradigm differs depending on the specific technical regime designated as Schumpeter Mark I or Mark II. In a Schumpeter Mark I regime, new entrants only apply new paradigms, whereas, in Mark II, assimilation is dependent on enterprises' absorptive capacity. Public research organizations are the source of radical discoveries and post-cooperation proposals aimed at companies with varying levels of technological maturity. Small and freshly founded entrepreneurial enterprises are more effective at disseminating the new paradigm in a Mark I regime. Still, the national research lab is supposed to target larger businesses in a Mark II system. The debate arises as to whether a national research lab program will perform worse if technology diffusion predominantly depends on small and medium-sized enterprises (SMEs). Policy measures must be aligned with the technology regime and the fitness of potential adopters to achieve successful diffusion and long-term effects in the changing technological landscape. The model uses a probabilistic learning technique to capture the distinctive characteristics of Mark I and Mark II situations while reflecting the inherent uncertainty in organizations' adoption of new technological paradigms.

$$Y_{i,t}^P = 1 - e^{-\varphi abs_{i,t}}; \varphi > 0 \quad (38)$$

The calculation of absorptive capacity involves aggregating the cumulative Research and Development (R&D) effort expended by firms, as outlined by Dose et al. (2021):

$$abs_{i,t} = \sum_{k=t_0}^{k=t-1} L_{i,k}^{RD} \quad (39)$$

Here, t_0 denotes the entrance historical of the firm into the market. In the context of the Mark II regime, incumbents exhibit a higher likelihood of harnessing the benefits of the new technological paradigm compared to new entrants. This distinction underscores the advantage that established firms possess in adopting and leveraging the innovative elements introduced by the emerging paradigm.

4.3.2. Computational findings

In a rigorous 100-iteration scenario study spanning 2020 to 2080, each policy is contrasted to a "no policy" baseline (see Table 6). Uniform magnitudes, set at 1 % of nominal GDP in scenarios III and IV, allow for accurate comparisons. Endogenous experiments I and II use a thorough trial-and-error method to achieve a consistent 1 % policy magnitude. Notably, policy experiment I uses trend = 0.13. The study investigates trade-offs between macroeconomic growth, public budget sustainability, and energy efficiency, categorizing the findings as "Policy-induced Improvement." Beginning with policy scenario I, an energy tax effectively increases energy efficiency by 20 %, but with a moderate success rate and no macroeconomic rebound. Rising energy costs contribute to higher unemployment, demanding energy tax collections and putting pressure on governmental finances.

Policy scenario II, which focuses on rewarding energy-efficient equipment, results in a 20 % increase in energy efficiency growth, comparable to the impact of an energy tax. This promotes specialized manufacturing, reduces energy consumption, and lowers unemployment while slightly increasing the deficit/GDP ratio. In contrast, scenario III's R&D subsidies boost the economy but only improve energy efficiency when used strategically (III - bis). The mission-oriented method in a Schumpeter Mark II environment excels, doubling or tripling growth rates, making significant energy efficiency gains, and demonstrating self-financing through unemployment benefit savings. Despite more substantial energy efficiency development, the Mark II regime only consistently results in higher GDP growth due to a productivity trade-off that needs a macroeconomic rebound impact.

4.4. Supplementary results

4.4.1. Policy shortfalls: an analysis of the consequences when strategies fall short

The investigation reveals a wide range of success rates (13 %–48 %) for environmental regulations, underscoring the complexities involved in their implementation. Interestingly, policy failures are more prevalent in standard scenarios characterized by low energy intensity, strong economic growth, and low unemployment. This suggests that policy inadequacies are more likely to occur under favorable economic and environmental conditions, highlighting the intricate relationship between economic performance and policy effectiveness.

Further analysis of policy outcomes complicates the assumption that certain regulations inherently provide "by-product" macroeconomic benefits. These benefits only materialize in scenarios where the policies succeed. In cases of failure, the anticipated advantages are either reversed or significantly diminished. This nuanced perspective underscores the need for a comprehensive evaluation that considers the complex interactions between macroeconomic factors, policy effectiveness, and environmental objectives.

The ratio between average marginal costs (MC) from failure and success sub-samples within baseline conditions appears in Table 8). A Wilcoxon test determines statistical significance through stars because the p-values measure differences between these ratios; ***** indicates p-values less than 0.01** while <0.10 indicates p-values less than 0.10.

4.4.2. Harmonizing strategies: navigating the complex landscape of policy mix

To accomplish long-term objectives, governments often combine several programs. In our simulations, we combined Policy I with Policies II, III, and IV and found that combining policies works better than a single policy. It is noteworthy that combining incentives for energy-efficient machinery with an energy tax produced a positive synergy that increased energy efficiency by more than 50 % without negatively impacting the economy. This emphasizes how a well-balanced policy mix can improve overall efficacy in achieving energy-saving targets. (See Table 9)

On the other hand, simulation results show a more complex link. For example, a policy combination with an increased energy tax rate and incentive programs can really improve energy efficiency a great deal. But doing so comes at the expense of economic growth, which noticeably slows down when this combination of policies is put into place. As such, in this policy framework, there is a trade-off between GDP growth and energy efficiency. (See Table 10)

When we investigate several policy combinations incorporating Policy I, we see only modestly favorable effects on unemployment and GDP growth. However, in most of these situations, there is a constant and statistically significant decrease in the shortfall/GDP ratio. This implies that the fiscal sustainability of public budgets dramatically depends on these particular policy combinations. The considerable reduction in the deficit/GDP ratio highlights the efficacy of various policy combinations in supporting the fiscal soundness of government budgets, even though their effects on GDP and unemployment may be negligible.

4.4.3. Navigating indirect approaches: balancing labor productivity and energy efficiency in policy

Our findings demonstrate the effectiveness of a high-incentive government spending program in improving energy efficiency. However, this triumph comes at a cost: economic growth would decelerate by more than 10 % associated to the baseline scenario. Increased incentives encourage energy efficiency improvements but at the expense of productivity, highlighting the difficulty of combining economic expansion and energy efficiency increases. Furthermore, the deficit/GDP ratio rises, demonstrating the enormous financial cost of attempting considerable energy reduction through indirect programs.

4.4.4. Beyond lab walls: investigating spillover effects in national research laboratories

Our findings demonstrate the effectiveness of a high-incentive government spending program in improving energy efficiency. However, this triumph comes at a cost: economic growth would decelerate by more than 10 % compared to the baseline scenario. Increased incentives encourage energy efficiency improvements but at the expense of productivity, highlighting the difficulty of combining economic expansion and energy efficiency increases. Furthermore, the deficit/GDP ratio rises, demonstrating the enormous financial cost of attempting considerable energy reduction through indirect programs.

$$\begin{cases} \xi 1_{p=l}^{EF} = \xi 1^{EF} + \theta_{p=l,t}^{EF}; \xi 2_{p=l}^{EF} = \xi 2^{EF} + \theta_{p=l,t}^{EF} \\ \xi 1_{p=l}^L = \xi 1^L + \zeta \theta_{p=l,t}^{EF}; \xi 2_{p=l}^L = \xi 2^L + \zeta \theta_{p=l,t}^{EF} \end{cases} \quad (40)$$

The outcomes of the nationwide lab trial are shown in the last section of (Table 5, which accounts for spillovers at two distinct rates: 25 % and 15 %. The national lab policy has a positive and statistically significant effect on the GDP growth rate when spillovers are large, outpacing the baseline scenario by about 4 %. Concurrently, this improvement helps to lower the deficit and unemployment rate.

Although a macroeconomic growth rebound impact is due to the faster GDP growth, it is not nearly as large as the gains from the policy-induced increases in energy efficiency. Remarkably, spillovers also contribute to the progress made in energy efficiency because the national lab strategy has improved the economic climate. In conclusion, adding spillovers makes the nationwide study lab policy a win-win approach that achieves energy savings without jeopardizing jobs or growth. Furthermore, this approach shows to be completely self-financing and does not put a burden on the public coffers.

4.4.5. Temporal perspectives: evaluating indirect policies versus national research labs across time horizons

Policies to improve energy efficiency have been assessed from 2020 to 2080, but political backing is more likely in shorter timeframes. The indirect policy combination (II + I) is more successful than the national research lab in 2030. The national lab will outperform other labs in macroeconomic results and energy efficiency by 2050.

5. Conclusion and policy implication

The investigation examines how various energy efficiency policies work using national economy agent-based models (ABMs) that implement direct and indirect technology policies. Integrating these methods produces innovative insights about innovation outputs generated by R&D activities from capital goods manufacturers and mission-driven national research labs.

This research evaluates opposing state approaches between entrepreneurial government intervention and standard market-correcting techniques that combine energy taxes and R&D funding. Our study demonstrates that tax-based energy policies, along with R&D subsidies, create fast technological uptake, but mission-oriented national research

Table 6

The table displays the Granger causality test findings obtained using [55] technique. P-values calculated with various lag selections using the Akaike Information Criterion (AIC) are represented by symbols such as ***, **, and * for significant levels. These results are based on 100 Monte Carlo simulations, which are summed annually.

	1 Lag	2 Lags
Energy Intensity → GDP growth	0.045**	0.003***
GDP growth → Energy Intensity	0.000***	0.00***

Table 5

Monte Carlo insights: A deep dive into correlation structures in BPF (6,32,12).

Series (bpf)	GDP (bpf)								
=	t-2	t-4	t + 1	t-1	t	t + 3	t + 2	t-3	t + 4
Energy demand	0.386	-0.166	0.6745	0.680	0.820	0.194	0.497	0.064	-0.042
Falling rate	-0.205	0.159	-0.600	-0.460	-0.619	-0.129	-0.406	0.029	0.111
Nominal wage	0.149	-0.098	0.264	0.243	0.279	0.183	0.222	0.022	0.164
Inflation	0.124	-0.284	0.164	0.273	0.277	-0.048	0.025	-0.103	-0.028
Unemployment	-0.595	0.017	-0.638	-0.797	-0.810	-0.122	-0.371	-0.283	0.032
Priv. investment	0.061	-0.280	0.349	0.247	0.389	0.205	0.280	-0.140	0.169
Priv. consumption	0.502	-0.153	0.885	0.827	0.975	0.257	0.604	0.131	-0.022
GDP	-0.111	0.174		0.549	0.872	1.000	0.872	0.549	0.174
									-0.111

Table 7

The (Table 7) compares the experiment performance to the baseline (no policy). Stars represent statistical significance (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The failure rate represents the proportion of simulations in which the energy strength in the strategy scenario does not fall under the zero. The macroeconomic rebound effect ("Rebound") is calculated using 100 Monte Carlo simulations between 2020 and 2080.

Policy	Energy efficiency growth	Energy intensity growth	Energy intensity at 2080	Energy demand growth	GDP growth	Unkempt.	Deficit/GDP	Failure rate	Rebound
Baseline	0.008	−0.007	4.243	0.018	0.024	0.142	0.019		
I	1.20**	1.20*	0.93	0.87*	0.98	1.09	0.82	40 %	−3 %
II	1.17*	1.25*	0.91	0.87	1.02	0.94	1.05	41 %	4 %
III	1.02	0.99	0.99	1.16***	1.18***	0.92**	0.90	48 %	>100 %
III - bis	1.42***	1.45**	0.88**	0.96	1.08***	0.84**	0.64	37 %	59 %
IV - Mark II	3.74***	3.98***	0.46***	0.04***	1.05***	0.65***	−0.47***	9 %	10 %
IV - Mark I	2.29***	2.39***	0.65***	0.57***	1.04***	0.82***	0.45**	17 %	4 %
Spillover = 25 %:									
IV - Mark II	3.50***	3.91***	0.44***	0.02***	1.01	0.65***	−0.13***	13 %	16 %
IV - Mark I	2.19***	2.37***	0.67***	0.55***	1.02	0.86***	0.53**	28 %	7 %
II ($q = 0.35$)	2.53***	3.51***	0.58***	−0.10***	0.85***	0.93	3.61***	21 %	−21 %
Spillover = 15 %:									
II + I ($tr^{en} = q = 0.25$)	2.32***	2.87***	0.59***	0.20***	0.93***	1.00	1.65***	23 %	−11 %
III - bis + I	1.66***	1.75***	0.76***	0.87	1.08***	0.97	0.61	34 %	56 %
IV - Mark II + I	3.37***	3.71***	0.42***	0.04***	0.99	0.75***	−0.42***	9 %	−2 %
II + I	1.60***	1.87***	0.75***	0.68***	1.00	0.98	0.67*	36 %	−2 %
IV - Mark II	3.42***	3.92***	0.41***	−0.03***	0.99	0.67***	−0.03***	13 %	0 %
IV - Mark I	2.19***	2.17***	0.72***	0.61***	1.02	0.89**	0.71	31 %	4 %
IV - Mark I + I	2.08***	2.24***	0.68***	0.61***	1.01	0.93	0.38**	17 %	1 %
III + I	1.41***	1.44*	0.85**	1.10	1.17***	0.95	0.48**	39 %	>100 %

Table 8

Policy	Energy int. in 2080	GDP growth	Unkempt.
IV - Mark II	0.59***	1.06**	0.76*
IV - Mark I	0.66***	1.05**	0.77**
III - bis	0.65***	1.00	0.83**
III	0.64***	1.00	0.68***
II	0.64***	1.02	0.80**
I	0.67***	1.04	0.75**

labs establish enduring economic advantages by achieving self-funded capabilities and boosting GDP growth through job creation. These policies need proper implementation time yet generate enduring sustainable advantages.

The research demonstrates why it is essential to handle energy efficiency and measure rebound effects. The effects of these policy measures produce minor macroeconomic growth-driven rebounds, which policy frameworks can successfully control. Short-term policymakers should implement tax-based interventions but simultaneously build long-term innovation investments to maintain the sustainability of energy

Table 9

Policy effects are presented based on success and failure scenarios. '+' indicates a positive effect, '(−)' denotes a negative effect, both with a meaning level of at least 5 %; = signals an inconsequential effect (significance level above 5 %). This concise notation enables a quick understanding of policy impacts on variables in different scenarios.

Policy	Successful runs			Failure runs		
	G.D.P growth	Unkempt.	Deficit/G.D.P	G.D.P growth	Unkempt.	Deficit/G.D.P
IV - Mark II	=	−	−	−	=	=
IV - Mark I	=	−	−	=	+	+
III - bias	+	−	−	+	+	+
III	+	−	−	+	+	+
I	=	−	−	=	+	+
II	=	−	−	=	+	+

Table 10

The table compares each experiment's average performance to the baseline (no policy) during three assessment stages: 2020–2030, 2020–2050, and 2020–2080. The indirect strategy is referred to as scenario II (II + I), while scenario IV - Mark II represents the national research lab. Stars indicate the meaning of the difference from the baseline, evaluated by a Wilcoxon test: *** for p -value < 0.01 , ** for p -value < 0.05 , and * for p -value < 0.10 .

Policy	Energy efficiency development	Energy strength growth	En. int. at evil. period	Energy demand development	GDP development	Unkempt.	Deficit/GDP	Failure rate
State study lab	4.31***	4.81***	0.51***	−0.04***	0.98	0.78***	−0.04***	14 %
Indirect policy 2080:	1.60***	1.87***	0.75***	0.68***	1.00	0.98	0.67*	36 %
Indirect policy 2030:	1.74***	1.90***	0.83***	0.57***	0.99	1.02	0.85	35 %
National research lab 2050:	3.16***	3.04***	0.67***	0.11***	1.02	0.84***	0.28***	23 %
National research lab	1.34	1.28	0.98	0.94	1.01	0.98	1.39*	48 %
Indirect policy	1.68**	2.01***	0.92***	0.47***	1.03	1.06	1.08	37 %

efficiency strategies.

5.1. Policy implications

The research findings generate essential guidelines that different stakeholders should implement. Governments' dual policy approach should include short-term energy taxation and sustained long-term investments in research institutions and technology adoption initiatives. The combined approach will create the best conditions for energy efficiency alongside stable economics and sustainable practices. The energy transition needs R&D subsidies to function as its fundamental policy instrument. Government institutions should dedicate more resources to public research facilities and national research laboratories to accelerate energy efficiency technology development rather than maintaining exclusive direct tax incentives.

Organizations need tailored policies that account for their unique energy consumption behaviours, technological development levels, and funding capacities. High-energy-intensive sectors would receive maximum benefits from specific tax incentives, but low-energy-intensive industries need regulatory support and market-based approaches. Government institutions must implement progressive taxation systems and establish precise efficiency targets to control possible increases in energy use stemming from improved energy efficiency. The implementation of targeted policies will reduce the adverse effects that energy savings create for total consumption levels.

Energy efficiency policies require integration into comprehensive climate action plans as their final step. Such policies need to transcend energy reduction measures to help achieve decarbonization goals and meet worldwide sustainability requirements through a comprehensive climate change strategy.

5.2. Limitations of the study

While the study generates valuable insights, there are several limitations to consider:

- **Agent-Based Modeling Limitations:** Although ABMs effectively simulate market dynamics, they may not fully capture real-world complexities such as behavioral uncertainties, geopolitical risks, and external energy price fluctuations. This limits the model's ability to predict long-term changes in energy efficiency outcomes accurately.
- **Data Constraints:** The forecasting model relies on historical data that faces three main issues: inconsistent data, outdated measurement methods, and a sparse coverage area. Insufficient data quality compromises the accuracy of long-term predictions, failing to capture modern global energy trends.
- **Social and Political Factors:** This model fails to include political and social factors together with cultural elements that impact policy execution because it does not consider public opinion, policy enforcement systems, or political stability. This analysis does not include the essential factors that determine the success of energy efficiency policies.
- **Geographic and Sectoral Generalizability:** The research results cannot be easily transferred between various industries and geographic regions. Different geographical regions and sectors display unique energy policy frameworks and infrastructure constraints that require additional research to develop effective policies.

5.3. Future research directions

Future research should address the following areas to strengthen the understanding of energy efficiency policies:

- **Improved Agent-Based Models:** Incorporating more diverse agent behaviors, external factors, and stochastic elements in ABMs will

enhance their ability to simulate real-world economic and energy system uncertainties. Integrating real-time data and predictive analytics will increase the accuracy of policy simulations and enable better evaluations of emerging technologies.

- **Cross-Industry and Cross-Region Comparisons:** Expanding the focus to compare policy effects across different industries and regions will provide a broader understanding of how energy efficiency policies perform under varied economic contexts. This should include assessing the interactions between public understanding, political systems, and the enforcement mechanisms that shape the success of energy efficiency measures.
- **Long-Term Policy Impact Analysis:** Future studies should evaluate the effects of energy efficiency policies over extended time horizons, taking into account structural economic changes and the diffusion of technological innovations. This will help policymakers understand how energy policies evolve over time and adapt to changing societal needs.

5.4. Final remarks

This study contributes to the ongoing discussion on the effectiveness of energy efficiency policies, emphasizing the need for long-term investments in innovation and sector-specific policy strategies. By integrating agent-based modeling with real-world policy considerations, our research provides a comprehensive framework for designing sustainable energy transitions. Further refinement of these models, coupled with expanded empirical analysis, will provide policymakers with the tools needed to develop adaptive, forward-thinking energy strategies for the future.

Credit author Statement

We, the undersigned authors of the manuscript titled “**Agent-Based Modeling of Energy Efficiency Policies: Enhancing Sustainable Development and Economic Growth**” hereby declare and affirm our individual contributions to the work as detailed below:

1Prof. Muhammad Mohsin * (corresponding author) led the project administration and supervision, and wrote the original draft of the manuscript.

2Chuanzhi Huo contributed to the conception and design of the study, as well as the interpretation of data.

3Manal Alidarous was responsible for data collection and analysis. He played a key role in data curation and formal analysis.

We confirm that all authors listed have made a significant contribution to the work, and have reviewed and approved the manuscript for submission.

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All authors have reviewed and approved the revised manuscript. The changes reflect our collective efforts to address the reviewers' comments and align the manuscript with the journal's requirements. We confirm that the removed author has been informed and has agreed to the change, and the new authors have provided their consent to be listed.

Declaration of competing interest

The authors assert that they have no identifiable competing financial interests or personal relationships that might have influenced the work reported in this paper.

Data availability

Data will be made available on request.

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